

## 6. Factorisation of Algebraic expressions

$$\begin{aligned} \textcircled{1} \quad & x^2 + 9x + 18 \\ \Rightarrow & = x^2 + 3x + 6x + 18 \\ & = x(x+3) + 6(x+3) \\ & = (x+3)(x+6) \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & y^2 + 24y + 144 \\ & = y^2 + 12y + 12y + 144 \\ & = y(y+12) + 12(y+12) \\ & = (y+12)(y+12) \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad & p^2 - 2p - 35 \\ & = p^2 - 7p + 5p - 35 \\ & = p(p-7) + 5(p-7) \\ & = (p-7)(p+5) \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad & m^2 - 23m + 120 \\ & = m^2 - 15m + 8m + 120 \\ & = m(m-15) + 8(m-15) \\ & = (m-15)(m+8) \end{aligned}$$

$$\begin{aligned} \textcircled{5} \quad & 2x^2 + x - 45 \\ & = 2x^2 + 9x + 5x - 45 \\ & = 2x(x+9) + 5(x+9) \\ & = (x+9)(2x+5) \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad & 44^2 - x - 3 \\ \rightarrow & 44^2 - 11x - 12x - 3 \\ & = 11(4-x) - 3(4-x) \\ & = (4-x)(11-3) \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad & x^2 - 10x + 9 \\ & = x^2 - 9x - 1x + 9 \\ & = x(x-9) - 1(x-9) \\ & = (x-9)(x-1) \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad & 5y^2 + 5y - 10 \\ & = y^2 + 5y + 10y - 10 \\ & = y(y+5) + 10(y+5) \\ & = (y+5)(y+10) \end{aligned}$$

$$\begin{aligned} \textcircled{9} \quad & p^2 - 7p - 44 \\ & = p^2 - 4p - 11p - 44 \\ & = p(p-4) - 11(p-4) \\ & = (p-4)(p-11) \end{aligned}$$

$$\begin{aligned} \textcircled{10} \quad & m^2 - 25m + 100 \\ & = m^2 - 20m + 5m + 100 \\ & = m(m-20) + 5(m-20) \\ & = (m-20)(m+5) \end{aligned}$$

$$\begin{aligned} \textcircled{11} \quad & 3x^2 + 14x + 15 \\ & = 3x^2 + 9x + 5x + 15 \\ & = 3x(x+3) + 5(x+3) \\ & = (x+3)(3x+5) \end{aligned}$$

$$\begin{aligned} \textcircled{11} \quad & 20x^2 - 26x + 8 \\ &= 20x^2 - 16x - 10x + 8 \\ &= 4x(5x-4) - 2(5x-4) \\ &= (5x-4)(4x-2) \end{aligned}$$

$$\left\{ \begin{aligned} 20 \times 8 &= 160 = 16 \times 10 \\ 20 \times (-2) &= -40 = -16 \times 2.5 \\ (-16 \times 10) + (-16 \times 2.5) &= -16(10+2.5) = -16(12.5) = -200 \end{aligned} \right.$$

Factors of  $a^3 + b^3$  :-

$$a^3 + b^3 = (a+b)(a^2 - ab + b^2)$$

Practice set 6.2

1. Factorise.

$$\begin{aligned} \textcircled{1} \quad & x^3 + 64y^3 = x^3 + (4y)^3 \\ \rightarrow & = (x+4y)(x^2 - x \cdot 4y + (4y)^2) \\ & = (x+4y)(x^2 - 4xy + 16y^2) \end{aligned}$$

$$\begin{aligned} \textcircled{2} \quad & 24a^3 + 81b^3 = 3(8a^3 + 27b^3) \\ \rightarrow & = 3[(2a)^3 + (3b)^3] \\ & = 3(2a+3b)[(2a)^2 - (2a \times 3b) + (3b)^2] \\ & = 3(2a+3b)(4a^2 - 6ab + 9b^2) \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad & y^3 + \frac{1}{8y^3} = y^3 + \frac{1}{(2y)^3} \\ & = \left(y + \frac{1}{2y}\right) \left[y^2 - y \times \frac{1}{2y} + \left(\frac{1}{2y}\right)^2\right] \\ & = \left(y + \frac{1}{2y}\right) \left(y^2 - \frac{1}{2} + \frac{1}{4y^2}\right) \end{aligned}$$



$$\begin{aligned} \textcircled{2} \quad 125p^3 + q^3 &= (5p)^3 + q^3 \\ &= (5p+q) \left[ (5p)^2 - 5p \times q + q^2 \right] \\ &= (5p+q) [25p^2 - 5pq + q^2] \end{aligned}$$

$$\begin{aligned} \textcircled{3} \quad 125k^3 + 27m^3 &= (5k)^3 + (3m)^3 \\ \rightarrow &= (5k+3m) \left[ (5k)^2 - 5k \times 3m + (3m)^2 \right] \\ &= (5k+3m) [25k^2 - 15km + 9m^2] \end{aligned}$$

$$\begin{aligned} \textcircled{4} \quad 2l^3 + 432m^3 &= 2(1l^3 + 216m^3) \\ \rightarrow &= 2(1l^3 + 6m^3) \\ &= 2(1l+6m) \left[ 1l^2 - 1l \times 6m + (6m)^2 \right] \\ &= 2(1l+6m) [1l^2 - 6lm + 36m^2] \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad a^3 + \frac{8}{a^3} &= \\ \rightarrow \quad a^3 + \left[ \frac{2}{a} \right]^3 &= \left( a + \frac{2}{a} \right) \left[ a^2 - a \times \frac{2}{a} + \left( \frac{2}{a} \right)^2 \right] \\ &= \left( a + \frac{2}{a} \right) \left[ a^2 - 2 + \frac{4}{a^2} \right] \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad 1 + \frac{q^3}{125} &= 1^3 + \left[ \frac{q}{5} \right]^3 \\ \rightarrow &= \left( 1 + \frac{q}{5} \right) \left[ 1^2 - 1 \times \frac{q}{5} + \left[ \frac{q}{5} \right]^2 \right] \\ &= \left( 1 + \frac{q}{5} \right) \left[ 1 - \frac{q}{5} + \frac{q^2}{25} \right] \end{aligned}$$

# Practice Set 6.3

$$\textcircled{1} a^3 - b^3 = (a-b)(a^2 + ab + b^2)$$

1. Factorise.

$$\begin{aligned} \textcircled{1} y^3 - 27 &= y^3 - 3^3 \\ \rightarrow &= (y-3)(y^2 + y \times 3 + 3^2) \\ &= (y-3)(y^2 + 3y + 9) \end{aligned}$$

$$\begin{aligned} \textcircled{2} x^3 - 64y^3 &= x^3 - (4y)^3 \\ &= (x-4y)(x^2 + x \times 4y + (4y)^2) \\ &= (x-4y)(x^2 + 4xy + 16y^2) \end{aligned}$$

$$\begin{aligned} \textcircled{3} 27m^3 - 216n^3 &= (3m)^3 - (6n)^3 \\ &= (3m-6n)(3m^2 + 3m \times 6n + (6n)^2) \\ &= (3m-6n)(9m^2 + 18mn + 36n^2) \end{aligned}$$

$$\begin{aligned} \textcircled{4} 125y^3 - 1 &= (5y)^3 - 1^3 \\ &= (5y-1)(5y^2 + 5y \times 1 + 1^2) \\ &= (5y-1)(25y^2 + 5y + 1) \end{aligned}$$

$$\begin{aligned} \textcircled{5} 8p^3 - \frac{27}{p^3} &= (2p)^3 - \left(\frac{3}{p}\right)^3 \\ &= \left(2p - \frac{3}{p}\right) \left[ (2p)^2 + 2p \times \frac{3}{p} + \left[\frac{3}{p}\right]^2 \right] \\ &= \left(2p - \frac{3}{p}\right) \left[ 4p^2 + 6 + \frac{9}{p} \right] \end{aligned}$$

$$\begin{aligned} \textcircled{6} \quad 343a^3 - 512b^3 &= (7a)^3 - (8b)^3 \\ &= (7a - 8b) [(7a)^2 + 7a \times 8b + (8b)^2] \\ &= (7a - 8b) [49a^2 + 56ab + 64b^2] \end{aligned}$$

$$\begin{aligned} \textcircled{7} \quad 64x^3 - 729y^3 &= (4x)^3 - (9y)^3 \\ &= (4x - 9y) [(4x)^2 + 4x \times 9y + (9y)^2] \\ &= (4x - 9y) [16x^2 + 36xy + 81y^2] \end{aligned}$$

$$\begin{aligned} \textcircled{8} \quad 16a^3 - \frac{128}{b^3} &= 16 \left( a^3 - \frac{8}{b^3} \right) \\ &= 16 \left( a^3 - \left( \frac{2}{b} \right)^3 \right) \\ &= 16 \left( a + \frac{2}{b} \right) \left[ a^2 + a \times \frac{2}{b} + \left( \frac{2}{b} \right)^2 \right] \\ &= 16 \left( a + \frac{2}{b} \right) \left[ a^2 + \frac{2a}{b} + \frac{4}{b^2} \right] \end{aligned}$$

2. Simplify :

$$\begin{aligned} \textcircled{1} \quad (x+y)^3 - (x-y)^3 &= x+y - (x-y) [(x+y)^2 + (x-y) + (x-y)^2] \\ &= x+y - x+y [x^2 + 2xy + y^2 + (x^2 - y^2) + (x^2 - 2xy + y^2)] \\ &= 2y [x^2 + 2xy + y^2 + x^2 - y^2 + x^2 - 2xy + y^2] \\ &= 2y [x^2 + x^2 + x^2 + y^2 - y^2 + y^2 + 2xy - 2xy] \end{aligned}$$



②  $(3a+5b)^3 - (3a-5b)^3$   
 $\rightarrow = (3xy-2ab) - (3xy+2ab) [(3xy+2ab)^2 + (3xy+2ab)(3xy-2ab) + (3xy-2ab)^2]$   
 $= 3xy-2ab - 3xy+2ab [(3xy)^2 + 2(3xy)(2ab) + (2ab)^2 + (3xy)^2 - 2(3xy)(2ab) + (2ab)^2]$   
 $= 3xy-2ab - 3xy+2ab [9x^2y^2 + 12xyab + 4a^2b^2 + 9x^2y^2 - 12xyab + 4a^2b^2]$   
 $= 3xy-2ab - 3xy+2ab [18x^2y^2 + 8a^2b^2]$   
 $= 3xy-2ab - 3xy+2ab [2(9x^2y^2 + 4a^2b^2)]$   
 $= 3xy-2ab - 3xy+2ab [2(9x^2y^2 + 4a^2b^2)]$

$30 \times \frac{10}{10} = 30$   
 $\frac{30}{10} = 3$   
 $3 \times 4 = 12$

$\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$   
 $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

$(x^2+y^2+z^2) + (x^2+y^2+z^2) + (x^2+y^2+z^2) = 3(x^2+y^2+z^2)$   
 $(x^2+y^2+z^2) + (x^2+y^2+z^2) + (x^2+y^2+z^2) = 3(x^2+y^2+z^2)$   
 $(x^2+y^2+z^2) + (x^2+y^2+z^2) + (x^2+y^2+z^2) = 3(x^2+y^2+z^2)$   
 $(x^2+y^2+z^2) + (x^2+y^2+z^2) + (x^2+y^2+z^2) = 3(x^2+y^2+z^2)$

10 Division

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## Practice Set 6.4

1. Simplify

$$1) \frac{m^2 - n^2}{(m+n)^2} \times \frac{m^2 + mn + n^2}{m^3 - n^3}$$

$$\Rightarrow = \frac{(m+n)(m-n)}{(m+n)(m+n)} \times \frac{m^2 + mn + n^2}{(m-n)(m^2 + mn + n^2)}$$

$$= \frac{1}{m+n}$$

$$2) \frac{a^2 + 10a + 21}{a^2 + 6a - 7} \times \frac{a^2 - 1}{a + 3}$$

$$\Rightarrow = \frac{(a+3)(a+7)}{(a+7)(a-1)} \times \frac{a^2 - 1}{a + 3}$$

$$= \frac{(a+3)}{(a-1)} \times \frac{(a+1)(a-1)}{(a+3)}$$

$$= a + 1$$

$$3) \frac{8x^3 - 27y^3}{4x^2 - 9y^2}$$

$$\Rightarrow = \frac{(2x)^3 - (3y)^3}{(2x)^2 - (3y)^2}$$

$$= \frac{(2x - 3y)[(2x)^2 + (2x)(3y) + (3y)^2]}{(2x + 3y)(2x - 3y)}$$

$$= \frac{4x^2 + 6xy + 9y^2}{2x + 3y}$$

$$4) \frac{x^2 - 5x - 24}{(x+3)(x+8)} \times \frac{x^2 - 64}{(x-8)^2}$$

$$\Rightarrow = \frac{(x+3)(x-8)}{(x+3)(x-8)} \times \frac{x^2 - 8^2}{(x-8)(x-8)}$$

$$= \frac{(x+8)(x-8)}{(x+8)(x-8)}$$

$$= 1$$

$$5) \frac{3x^2 - x - 2}{x^2 - 7x + 12} \div \frac{3x^2 - 7x - 6}{x^2 - 4}$$

$$\Rightarrow = \frac{3x^2 - 3x - 2x - 2}{(x+3)(x-4)} \div \frac{3x^2 - 9x + 2x - 6}{x^2 - 2^2}$$

$$= \frac{3x(x-1) + 2(x-1)}{(x-3)(x-4)} \div \frac{3x(x-3) + 2(x-3)}{(x+2)(x-2)}$$

$$\therefore \frac{(x-1)(3x+2)}{(x-3)(x-4)} \times \frac{(x+2)(x-2)}{(x-3)(3x-2)}$$

$$= \frac{(x-1)(x+2)(x-2)}{(x+3)(x+4)(x+3)}$$

$$= \frac{(x+1)(x+2)(x+2)}{(x-4)(x+3)^2}$$

$$6) \frac{4x^2 - 19x + 6}{16x^2 - 9}$$

$$\Rightarrow = 4x - 8x - 3x + 6$$

$$= 4x(x-2) - 3(x-2)$$

$$= (x-2)(4x-3)$$

$$16x^2 - 9 = (4x)^2 (3)^2$$

$$= \frac{(x-2)(4x-3)}{(4x+3)(4x-3)}$$

$$= \frac{(x+2)}{(4x+3)}$$

$$\left[ \begin{array}{l} 4 \times 6 = 24 \\ 3 \times 8 = 24 \\ -3 - 8 = -11 \end{array} \right]$$



$$7) \frac{a^3 - 27}{5a^2 - 16a + 3} \div \frac{a^2 + 3a + 9}{25a^2 - 1}$$

$$8) \frac{1 - 2x + x^2}{1 - x^3} \times \frac{1 + x + x^2}{1 + x}$$

$$\Rightarrow \frac{x^2 + 2x - 1}{1 - x^3} \times \frac{1 + x + x^2}{1 + x}$$

$$= \frac{(1+x)(1-x)}{(1-x)(1+x+x^2)} \times \frac{1+x+x^2}{1+x}$$

$$= \frac{1-x}{1+x}$$